

ABSTRACT

This research paper presents a comprehensive comparison of four pre-trained models for sentiment analysis applied to COVID-19 news headlines. The COVID-19 pandemic has triggered an unprecedented flow of information and public discourse, making it essential to understand the sentiments and opinions expressed in news headlines. The study evaluates the performance of RoBERTa, VADER, BERT, and TextBlob in this specific context.

In this research, a diverse dataset of COVID-19 news headlines is collected and preprocessed. Each of the four models is employed to perform sentiment analysis on this dataset. The evaluation involves assessing the accuracy, precision, recall, and F1 score of these models in capturing the nuances of sentiment and emotion within the headlines. The findings of this comparative analysis reveal important insights into the strengths and limitations of each model. RoBERTa, a powerful transformer model, exhibits a robust performance in understanding the subtleties of sentiment in COVID-19 news headlines. VADER, a rule-based system, demonstrates its adaptability to this domain. BERT, a sibling of RoBERTa, showcases its ability to discern sentiment nuances, albeit with some complexities. TextBlob, a simplistic rule-based library, provides a user-friendly approach to sentiment analysis.

This research paper is valuable for researchers, policymakers, and businesses seeking to gauge public sentiment during a global health crisis and make informed decisions based on this sentiment analysis. It not only compares these models but also offers insights into the practical use of sentiment analysis in real-world scenarios, enhancing our understanding of public opinion during the COVID-19 pandemic.

I. INTRODUCTION

In today's fast-paced digital age, news headlines serve as the first point of contact for readers seeking to understand the world around them. Newspaper headlines not only convey information but also have the power to shape

public perceptions and sentiments. The ability to analyze these sentiments effectively is of paramount importance, as it enables us to gauge public opinion, track evolving trends, and make informed decisions based on the collective mood of society.

Sentiment analysis plays a crucial role in understanding public sentiment and emotions surrounding specific events or topics. In the context of COVID-19, it is vital to gauge public sentiment towards the pandemic to identify trends, concerns, and public perceptions. In this comparison, we will evaluate the performance of four pre-trained models in sentiment analysis, namely RoBERTa, VADER, BERT, and TextBlob, when applied to COVID-19 news headlines.

1. RoBERTa:

RoBERTa, a variant of the BERT model, is a state-of-the-art natural language processing model known for its exceptional performance in various NLP tasks. It has been pre-trained on a vast corpus of text data, making it capable of understanding the nuances of language. We will examine how well RoBERTa can adapt to COVID-19 news headlines and accurately identify sentiments in these texts.

2. VADER (Valence Aware Dictionary and sentiment Reasoner):

VADER is a lexicon and rule-based sentiment analysis tool designed to assess sentiment in text data. It is specifically tailored for social media text but can be applied to news headlines as well. We will explore VADER's effectiveness in detecting sentiment within COVID-19 news headlines and whether it can provide meaningful insights into public opinion regarding the pandemic.

3. BERT (Bidirectional Encoder Representations from Transformers):

BERT is another powerful transformer-based model that has proven its worth in NLP tasks. We will assess how BERT, a sibling of RoBERTa, performs in sentiment analysis when exposed to COVID-19 news headlines. Its ability to understand context and semantics will be scrutinized.

4. TextBlob:

TextBlob is a simpler, rule-based sentiment analysis library that is often used for quick sentiment analysis tasks. It might not match the complexity of RoBERTa or BERT, but it offers a user-friendly, straightforward approach to sentiment analysis. We will examine its performance in assessing the sentiment of COVID-19 news headlines and whether it can provide valuable insights with minimal complexity.

In this comparison, we aim to understand the strengths and weaknesses of these four models in the context of COVID-19 news headlines sentiment analysis. By evaluating their accuracy, ability to handle domain-specific language, and their ease of use, we can provide insights into the best model for this specific task. The results will help researchers, businesses, and policymakers better comprehend public sentiment and make informed decisions during the ongoing pandemic.

II . LITERATURE SURVEY

The COVID-19 pandemic, which emerged in late 2019, has had an unprecedented impact on the world. Beyond its health implications, the pandemic has given rise to a flood of information and news coverage, with headlines often shaping public perceptions and reactions. Sentiment analysis, a subfield of natural language processing (NLP), plays a crucial role in understanding how people perceive and react to the ongoing pandemic. Sentiment analysis involves the use of computational techniques to determine the sentiment or emotional tone expressed in a piece of text, which can range from positive to negative or neutral. The purpose of this literature review is to compare and evaluate the performance of four pre-trained NLP models in sentiment analysis specifically applied to COVID-19 news headlines. The models under consideration are:

VADER (Valence Aware Dictionary and sentiment Reasoner): VADER is a lexicon and rule-based sentiment analysis tool designed for social media text. It is known for its simplicity and speed

VADER (Valence Aware Dictionary and sentiment Reasoner): VADER is a lexicon and rule.

BERT (Bidirectional Encoder Representations from Transformers): BERT is a transformer-based model that has achieved state-of-the-art performance in various NLP tasks, including sentiment analysis.

RoBERTa (A Robustly Optimized BERT Pretraining Approach): RoBERTa is an extension of BERT that further optimizes pre-training for better performance in downstream tasks.

TextBlob: TextBlob is a simpler and lightweight NLP library that includes a pre-trained sentiment analysis module. It is often used for basic sentiment analysis tasks.

This literature review will assess the performance of these models based on criteria such as accuracy, precision, recall, F1-score, and computational efficiency. Additionally, it will consider their ability to handle nuances in COVID-19-related sentiment, including the impact of rapidly changing information, evolving public attitudes, and the prevalence of medical jargon.

The research papers that have been taken into consideration during the course of research and study are different from the paper in the following ways:

[1] The referenced paper analyzes tweets using BERT and other machine learning methods. There are various advantages of pre-trained model over other machine learning methods such as Transfer learning, Reduced Data Requirements, Lower Development Time, Better generalization. The referenced paper only analyzes one pre-trained model that lacks by other three in community support, ease of use, resource efficiency, simplicity and speed.

[2] The referenced paper analyzes different models from Transformers and compares them on the basis of F1 score, MCC and accuracy score of the models on news headlines. The pre trained models that are used in this paper are best for real time application due to their speed and efficiency, they are preferred to be used for real time application like social media monitoring and

chatbots. In this paper, the news headlines that are specific to covid-19 are analyzed.

III . MODELS USED

In this, four distinct pre-trained models were utilized to analyze the sentiment of COVID-19 news headlines. These models encompass a range of methodologies and complexity levels:

1. RoBERTa:

- RoBERTa (A Robustly Optimized BERT Pretraining Approach) is a transformer-based model that builds upon the BERT architecture. It is known for its state-of-the-art performance in various natural language processing (NLP) tasks. RoBERTa has been pre-trained on a large corpus of text data, enabling it to understand the intricacies of language and context. In this study, RoBERTa was employed to assess its suitability for sentiment analysis on COVID-19 news headlines.

2. VADER (Valence Aware Dictionary and sentiment Reasoner):

- VADER is a lexicon and rule-based sentiment analysis tool designed to evaluate sentiment in text data. It uses a pre-built lexicon of words and a set of grammatical rules to determine the sentiment polarity of text. VADER is particularly useful for social media and short text analysis but can also be applied to longer texts, such as news headlines.

3. BERT (Bidirectional Encoder Representations from Transformers):

- BERT, like RoBERTa, is a transformer-based model used for various NLP tasks. BERT stands for Bidirectional Encoder Representations from Transformers, and it has demonstrated exceptional performance in understanding context and semantics in text. In this study, BERT was assessed for its capacity to identify and analyze sentiment in COVID-19 news headlines.
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4. TextBlob:

- TextBlob is a simple, rule-based sentiment analysis library that provides an easy-to-use approach for sentiment analysis. It relies on predefined rules and a sentiment lexicon to classify text as positive, negative, or neutral. While not as complex as transformer models like RoBERTa or BERT, TextBlob offers a straightforward way to perform sentiment analysis.

Each of these models was used to conduct sentiment analysis on a dataset of COVID-19 news headlines. The research paper compared their performance in terms of accuracy, precision, recall, and F1 score to determine their effectiveness in capturing the nuanced sentiments expressed in these headlines. This comprehensive evaluation allowed the paper to provide insights into the strengths and weaknesses of each model in the specific context of COVID-19 news headlines, assisting researchers, policymakers, and businesses in making informed decisions based on public sentiment.

IV. EXPERIMENTAL SETUP

We used a labeled dataset of newspaper articles, consisting of a diverse range of news topics, to train and evaluate the sentiment analysis models. The dataset was preprocessed to remove noise and irrelevant information, and then divided into training and testing sets.

For each model, we fine-tuned or configured the model based on the specific requirements of the sentiment analysis task. The models were then evaluated using common evaluation metrics, including the F1 score and accuracy.

Model Performance

Table 1 presents the F1 scores and accuracy percentages obtained by each model.

Table 1: Model Performance for Sentiment Analysis

percentages. The experiments were conducted on a dataset of newspaper articles to gauge the effectiveness of the models in analyzing sentiment within this specific domain.

Model	F1 Score	Accuracy
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RoBERTa	80.92%	87.85%
BERT	75.86%	83.00%
VADER	54.00%	54.00%
TextBlob	58.00%	60.00%

V. DISCUSSION

The experimental results illustrate that RoBERTa achieved the highest F1 score and accuracy, followed by BERT. Both transformer-based models demonstrated superior performance compared to the rule-based models, VADER and TextBlob, in terms of both F1 score and accuracy.

RoBERTa and BERT, being transformer-based models, excel at capturing contextual information effectively, allowing for more accurate sentiment analysis. Despite their complexity, these models demonstrated high accuracy in this domain. On the other hand, VADER and TextBlob, though simpler in design, provided reasonable accuracy and can be computationally efficient alternatives for sentiment analysis tasks.

In conclusion, the experimental results suggest that transformer-based models, particularly RoBERTa, show promise for sentiment analysis in the context of newspaper articles. Further research and fine-tuning of these models could potentially enhance their accuracy and broaden their applicability in real-world sentiment analysis tasks.

VI. RESULT

In this section, we present the results of our sentiment analysis experiments using four different models: RoBERTa, BERT, VADER, and TextBlob. The evaluation of these models was based on their F1 scores and accuracy

VII. REFERENCES

- Comparison of BERT Models and Machine Learning Methods for Sentiment Analysis on Turkish Tweets
 - [Zekeriya Anil Guven](#)
- Sentiment Analysis of News Headlines Using Simple Transformers
 - [Anurag Singh](#); [Goonjan Jain](#)
- Comparative Analysis of Transformer Based Pre-Trained NLP Models
 - [Saurav Singla](#)
- Sentiment Analysis Applied to News from the Brazilian Stock Market
 - Brenda A. Januário; Arthur Emanuel de O. Carosia; Ana Estela A. da Silva
- Headlines Articles Analysis and NLP
 - [Cristiane Silva](#) (Published in Towards Data Science)

